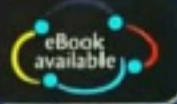


OXFORD
HIGHER EDUCATION

FREE

ONLINE RESOURCES
For Teachers and Students



MACHINE LEARNING

S. Sridhar | M. Vijayalakshmi



MACHINE LEARNING

Dr S. Sridhar

Professor,

*Department of Information Science and Technology,
College of Engineering, Guindy Campus,
Anna University, Chennai*

Dr M. Vijayalakshmi

Associate Professor,

*Department of Information Science and Technology,
College of Engineering, Guindy Campus,
Anna University, Chennai*

OXFORD
UNIVERSITY PRESS

OXFORD
UNIVERSITY PRESS

Oxford University Press is a department of the University of Oxford.
It furthers the University's objective of excellence in research, scholarship,
and education by publishing worldwide. Oxford is a registered trade mark of
Oxford University Press in the UK and in certain other countries.

Published in India by
Oxford University Press
22 Workspace, 2nd Floor, 1/22 Asaf Ali Road, New Delhi 110002

© Oxford University Press 2021

The moral rights of the author/s have been asserted.

First Edition published in 2021

All rights reserved. No part of this publication may be reproduced, stored in
a retrieval system, or transmitted, in any form or by any means, without the
prior permission in writing of Oxford University Press, or as expressly permitted
by law, by licence, or under terms agreed with the appropriate reprographics
rights organization. Enquiries concerning reproduction outside the scope of the
above should be sent to the Rights Department, Oxford University Press, at the
address above.

You must not circulate this work in any other form
and you must impose this same condition on any acquirer.

ISBN-13 (print edition): 978-0-19-012727-5
ISBN-10 (print edition): 0-19-012727-9

Typeset in Palatino Linotype
by B2K-Bytes 2 Knowledge, Tamil Nadu
Printed in India by

Cover image: © whiteMocca/Shutterstock

For product information and current price, please visit www.india.oup.com

Third-party website addresses mentioned in this book are provided
by Oxford University Press in good faith and for information only.
Oxford University Press disclaims any responsibility for the material contained therein.

Dedicated to my mother, late Mrs Parameswari Sundaramurthy, and my mother-in-law, late Mrs Renuga Nagarajan, whose encouragement and moral support motivated me to start writing this book when they were alive and remained an eternal motivation even after their demise in completing this book.

Dr S. Sridhar

Dedicated with love and affection to my family, Husband (Mathan), Sons (Sam and Ayden), and Mom (Vennila).

– Dr M. Vijayalakshmi

Oxford University Press

Preface

Can machines learn like human beings? This question has been posed over many decades and the search for an answer resulted in the domain of artificial intelligence. Informally, learning is nothing but adaptability to the environment. Human beings can adapt to the environment using the learning process. Machine Learning is a branch of artificial intelligence which is defined as “the field of study that gives computers the ability to learn without any explicit programming”. Unlike other user-defined programs, machine learning programs try to learn from data automatically. Initially, computer scientists and researchers used automatic learning through logical reasoning. However, much progress could not be made using logic. Eventually, machine learning became popular due to the success of data-driven learning algorithms.

Human beings have always considered themselves as the superior species in object recognition. While machines can crunch numbers in seconds, human beings have shown superiority in recognizing objects. However, recent applications in deep learning show that computers are also good in facial recognition. The recent developments in machine learning such as object recognition, social media analysis like sentiment analysis, recommendation systems including Amazon’s book recommendation, innovations, for example driverless cars, and voice assistance systems, which include Amazon’s Alexa, Microsoft’s Cortana and Google Assistant, have created more awareness about machine learning. The availability of smart phones, IoT, and cloud technology have brought these machine learning technologies to daily life.

Business and government organizations benefit from machine learning technology. These organizations traditionally have a huge amount of data. Social networks such as Twitter, YouTube, and Facebook generate data in terms of Terabytes (TB), Exabytes (EB), and Zettabytes (ZB). In technologies like IoT, sensors generate a huge amount of data independent of any human intervention. A sudden interest has been seen in using machine learning algorithms to extract knowledge from these data automatically. Why? The reason being that extracted knowledge can be useful for prediction and helps in better decision making. This facilitates the development of many knowledge-based and intelligent applications. Therefore, awareness of basic machine learning is a must for students and researchers, computer scientists and professionals, data analysts and data scientists. Historically, these organizations used statistics to analyze these data, but statistics could not be applied on big data. The need to process enormous amount of data poses a challenge as new techniques are required to process this voluminous data, and hence, machine learning is the driving force for many fields such as data science, data analysis, data analytics, data mining, and big data analytics.

Scope of the Book

Our aim has been to provide a first level textbook that follows a simple algorithmic approach and comes with numerical problems to explain the concepts of machine learning. This book stems from the experience of the authors in teaching the students at the Anna University and the National Institute of Technology for over three decades. It targets the undergraduate and post-graduate students in computer science, information technology, and in general, engineering students. This book is also useful for the ones who study data science, data analysis, data analytics, and data mining.

This book comprises chapters covering the concepts of machine learning, a laboratory manual that consists of 25 experiments implemented in Python, and appendices on the basics of Python and Python packages. The theory part includes many numerical problems, review questions and pedagogical support such as crossword and word search. Additional material is available as online resources for better understanding of the subject.

Key Features of the Book

- Uses only minimal mathematics to understand the machine learning algorithms covered in the book
- Follows an algorithmic approach to explain the basics of machine learning
- Comes with various numerical problems to emphasize on the important concepts of data analytics
- Includes a laboratory manual for implementing machine learning concepts in Python environment
- Has two appendices covering the basics of Python and Python packages
- Focuses on pedagogy like chapter-end review and numerical questions, crosswords and jumbled word searches
- Illustrates important and latest concepts like deep learning, regression analysis, support vector machines, clustering algorithms, and ensemble learning

Content and Organization

The book is divided into 16 chapters and three appendices. The appendices A, B, and C can be accessed through the QR codes provided in the table of contents.

Chapter 1 introduces the *Basic Concepts of Machine Learning* and explores its relationships with other domains. This chapter also explores the machine learning types and applications.

Chapter 2 of this book is about *Understanding Data*, which is crucial for data-driven machine learning algorithms. The mathematics that is necessary for understanding data such as linear algebra and statistics covering univariate, bivariate and multivariate statistics are introduced in this chapter. This chapter also includes feature engineering and dimensionality reduction techniques.

Chapter 3 covers the *Basic Concepts of Learning*. This chapter discusses about theoretical aspects of learning such as concept learning, version spaces, hypothesis, and hypothesis space. It also introduces learning frameworks like PAC learning, mistake bound model, and VC dimensions.

Chapter 4 is about *Similarity Learning*. It discusses instance-based learning, nearest-neighbor learning, weighted k-nearest algorithms, nearest centroid classifier, and locally weighted regression (LWR) algorithms.

Chapter 5 introduces the basics of *Regression*. The concepts of linear regression and non-linear regression are discussed in this chapter. It also covers logistic regression. Finally, this chapter outlines the recent algorithms like Ridge, Lasso, and Elastic Net regression.

Chapter 6 throws light on the concept of *Decision Tree Learning*. The concept of information theory, entropy, and information gain are discussed in this chapter. The basics of tree construction

algorithms like ID3, C4.5, CART, and Regression Trees and its illustration are included in this chapter. The decision tree evaluation is also introduced here.

Chapter 7 discusses *Rule-based Learning*. This chapter illustrates rule generation. The sequential covering algorithms like PRISM and FOIL are introduced here. This chapter also discusses analytical learning, explanation-based learning, and active learning mechanisms. An outline of association rule mining is also provided in this chapter.

Chapter 8 introduces the basics of *Bayesian* model. The chapter covers the concepts of classification using the Bayesian principle. Naïve Bayesian classifier and Continuous Features classification are introduced in this chapter. The variants of Bayesian classifier are also discussed.

Chapter 9 discusses *Probabilistic Graphical Models*. The discussion of the Bayesian Belief network construction and its inference mechanism are included in this chapter. Markov chain and Hidden Markov Model (HMM) are also introduced along with the associated algorithms.

Chapter 10 introduces the basics of *Artificial Neural Networks (ANN)*. The chapter introduces the concepts of neural networks such as neurons, activation functions, and ANN types. Perceptron, back-propagation neural networks, Radial Basis Function Neural Network (RBFNN), Self-Organizing Feature Map (SOFM) are covered here. The chapter ends with challenges and some applications of ANN.

Chapter 11 covers *Support Vector Machines (SVM)*. This chapter begins with a gentle introduction of linear discriminant analysis and then covers the concepts of SVM such as margins, kernels, and its associated optimization theory. The hard margin and soft margin SVMs are introduced here. Finally, this chapter ends with support vector regression.

Chapter 12 introduces *Ensemble Learning*. It covers meta-classifiers, the concept of voting, bootstrap resampling, bagging, and random forest and stacking algorithms. This chapter ends with the popular AdaBoost algorithm.

Chapter 13 discusses *Cluster Analysis*. Hierarchical clustering algorithms and partitional clustering algorithms like *k*-means algorithm are introduced in this chapter. In addition, the outline of density-based, grid-based and probability-based approaches like fuzzy clustering and EM algorithm is provided. This chapter ends with an evaluation of clustering algorithms.

Chapter 14 covers the concept of *Reinforcement Learning*. This chapter starts with the idea of reinforcement learning, multi-arm bandit problem and Markov Decision Process (MDP). It then introduces model-based (passive learning) and model-free methods. The Q-Learning and SARSA concepts are also covered in this chapter.

Chapter 15 is about *Genetic Algorithms*. The concepts of genetic algorithms and genetic algorithm components along with simple examples are present in this chapter. Evolutionary computation, like simulated annealing, and genetic programming are outlined at the end of this chapter.

Chapter 16 discusses *Deep Learning*. CNN and RNN are explained in this chapter. Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are outlined here. Additional web contents are provided for a thorough understanding of deep learning.

Appendix A discusses Python basics.

Appendix B covers Python packages that are necessary to implement the machine learning algorithms. The packages like Numpy, Pandas, Matplotlib, Scikit-learn and Keras are outlined in this appendix.

Appendix C offers 25 laboratory experiments covering the concepts of the textbook.

Acknowledgments

"The Lord is my strength and my shield; my heart trusted in him, and I am helped: therefore my heart greatly rejoiceth; and with my song will I praise him." [Psalm 28:7]

First and foremost, the authors express the gratitude with praises and thanks to **Lord God** who made this thing happen. This book would not have been possible without the help of many friends, students, colleagues, and well-wishers. The authors thank all the students who motivated them by asking them the right questions at appropriate time. The authors also thank the colleagues of the Department of Information Science and Technology, Anna University, SRM University, and National Institute of Technology for supporting them with their constant reviews and suggestions. The authors acknowledge the support of the reviewers for providing critical suggestions to improve the quality of the book.

We thank the editorial team of the Oxford University Press, India for their support in bringing the book to realization. The authors thank the members of the marketing team too for their continuous support and encouragement during the development of this book.

Dr Sridhar thanks his wife, Dr Vasanthi, and daughters, Dr Shobika and Shree Varshika, for being patient and accommodative. He also wants to recollect the contributions made by his mother, Mrs Parameswari Sundaramurthy, and mother-in-law, Mrs Renuga Nagarajan, who provided constant encouragement and remained an eternal motivation in completing the book.

Dr Vijayalakshmi would like to thank her husband, Mr Mathan, Sons, Sam and Ayden, and her mother, Ms Vennila, for supporting her in completing this book successfully.

Any comments and suggestions for further improvement of the book are welcome; please send them at ssridhar2004@gmail.com and his website www.drssridhar.com, or vijim@auist.net.

S. Sridhar
M. Vijayalakshmi

The Oxford University Press, India would like to thank all the reviewers:

1. Dr Shriram K. Vasudevan (K. Ramakrishnan College of Technology)
2. Maniroja M. Edinburgh (EXTC Dept., TSEC)
3. Nanda Dulal Jana (NIT, Durgapur)
4. Dipankar Dutta (NERIM Group of Institutions)
5. Imthias Ahamed T P (TKM College of Engineering)
6. Dinobandhu Bhandari (Heritage Institute of Technology)
7. Prof. BSP Mishra (KIIT, Patia)
8. Mrs Angeline (SRM University)
9. Dr Ram Mohan Dhara (IMT, Ghaziabad)
10. Saurabh Sharma (KIET Group of Institutions, Ghaziabad)
11. B. Lakshmanan (Mepco Schlenk Engg. College)
12. Dr T. Subha (Sairam Engineering College)
13. Dr R. Ramya Devi (Velammal Engineering College)
14. Mr Zatin Kumar (Raj Kumar Goel Institute of Technology, Ghaziabad)
15. M. Sowmiya (Easwari Engg. College)
16. A. Sekar (Knowledge Institute of Technology)

Oxford University Press. All rights reserved.

QR Code Content Details

Scan the QR codes provided under the sections listed below to access the following topics:

Table of Contents

- Appendix 1 – Python Language Fundamentals
- Appendix 2 – Python Packages
- Appendix 3 – Lab Manual (with 25 exercises)

Chapter 1

- Section 1.4.4 – Important Machine Learning Algorithms

Chapter 2

- Section 2.1 – Machine Learning and Importance of Linear Algebra, Matrices and Tensors, Sampling Techniques, Information Theory, Evaluation of Classifier Model and Additional Examples
- Section 2.5 – Measures of Frequency and Additional Examples
- Exercises – Additional Key Terms, Short Questions and Numerical Problems

Chapter 3

- Section 3.1 – A figure showing Types of Learning
- Section 3.4.4 – Additional Examples on Generalization and Specialization
- Section 3.4.6 – Additional Examples on Version Spaces

Chapter 5

- Section 5.3 – Additional Examples

Chapter 6

- Section 6.2.1 – Additional Examples

Chapter 7

- Section 7.1 – Simple OneR Algorithm
- Section 7.8 – Additional Examples
- Section 7.8 – Additional Examples: Measure 3-Lift
- Section 7.8 – Apriori Algorithm and Frequent Pattern (FP) Growth Algorithm
- Exercises – Additional Questions

Chapter 8

- Section 8.1 – Probability Theory and Additional Examples

Chapter 9

- Section 9.3.1 – Additional Examples

Chapter 10

- Section 10.1 – Convolution Neural Network, Modular Neural Network, and Recurrent Neural Network
- Key Terms – Additional Key Terms

Chapter 11

- Section 11.1 – Decision Functions and Additional Examples

Chapter 12

- Section 12.2 – Illustrations of Types of Ensemble Models, Parallel Ensemble Model, Voting, Bootstrap Resampling, Working Principle of Bagging, Learning Mechanism through Stacking, and Sequential Ensemble Model
- Section 12.4.1 – Gradient Tree Boosting and XGBoost
- Exercises – Additional Review Question

Chapter 13

- Section 13.2 – Additional Information on Proximity Measures
- Section 13.2 – Taxonomy of Clustering Algorithms
- Section 13.3.4 – Additional Examples
- Section 13.4 – Additional Examples
- Section 13.8 – Purity, Evaluation based on Ground Truth, and Similarity-based Measures
- Key Terms – Additional Key Terms
- Numerical Problems – Additional Numerical Problems

Chapter 14

- Section 14.1 – Additional Information
- Section 14.2 – Context of Reinforcement Learning and SoftMax Method
- Section 14.7 – Additional Information on Dynamic Programming
- Section 14.7 – Additional Examples

Chapter 15

- Section 15.2 – Examples of Optimization Problems and Search Spaces
- Section 15.3 – Flowchart of a Typical GA Algorithm
- Section 15.4.6 – Additional Examples and Information on Topics Discussed in Section 15.4
- Section 15.5.1 – Feature Selection using Genetic Algorithms and Additional Content on Genetic Algorithm Classifier
- Section 15.6.2 – Additional Information on Genetic Programming

Chapter 16

- Section 16.2 – Additional Information on Activation Functions
- Section 16.3 – $L1$ and $L2$ Regularizations
- Section 16.4 – Additional Information on Input Layer
- Section 16.4 – Additional Information on Padding
- Section 16.5 – Detailed Information on Transfer Learning
- Section 16.8 – Restricted Boltzmann Machines and Deep Belief Networks, Auto Encoders, and Deep Reinforcement Learning
- Exercises – Additional Questions

Detailed Contents

<i>Preface</i>	<i>iv</i>
<i>Acknowledgements</i>	<i>vii</i>
<i>QR Code Content Details</i>	<i>viii</i>

1. Introduction to Machine Learning	1	2.5.1 Data Visualization	36
1.1 Need for Machine Learning	1	2.5.2 Central Tendency	38
1.2 Machine Learning Explained	3	2.5.3 Dispersion	40
1.3 Machine Learning in Relation to Other Fields	5	2.5.4 Shape	41
1.3.1 Machine Learning and Artificial Intelligence	5	2.5.5 Special Univariate Plots	43
1.3.2 Machine Learning, Data Science, Data Mining, and Data Analytics	5	2.6 Bivariate Data and Multivariate Data	44
1.3.3 Machine Learning and Statistics	6	2.6.1 Bivariate Statistics	46
1.4 Types of Machine Learning	7	2.7 Multivariate Statistics	47
1.4.1 Supervised Learning	8	2.8 Essential Mathematics for Multivariate Data	49
1.4.2 Unsupervised Learning	11	2.8.1 Linear Systems and Gaussian Elimination for Multivariate Data	49
1.4.3 Semi-supervised Learning	12	2.8.2 Matrix Decompositions	50
1.4.4 Reinforcement Learning	12	2.8.3 Machine Learning and Importance of Probability and Statistics	52
1.5 Challenges of Machine Learning	13	2.9 Overview of Hypothesis	57
1.6 Machine Learning Process	14	2.9.1 Comparing Learning Methods	59
1.7 Machine Learning Applications	15	2.10 Feature Engineering and Dimensionality Reduction Techniques	62
2. Understanding Data	22	2.10.1 Stepwise Forward Selection	63
2.1 What is Data?	22	2.10.2 Stepwise Backward Elimination	63
2.1.1 Types of Data	24	2.10.3 Principal Component Analysis	63
2.1.2 Data Storage and Representation	25	2.10.4 Linear Discriminant Analysis	67
2.2 Big Data Analytics and Types of Analytics	26	2.10.5 Singular Value Decomposition	68
2.3 Big Data Analysis Framework	27		
2.3.1 Data Collection	29		
2.3.2 Data Preprocessing	30		
2.4 Descriptive Statistics	34		
2.5 Univariate Data Analysis and Visualization	36		

3. Basics of Learning Theory	77	4.2 Nearest-Neighbor Learning	117
3.1 Introduction to Learning and its Types	78	4.3 Weighted K-Nearest-Neighbor Algorithm	120
3.2 Introduction to Computation Learning Theory	80	4.4 Nearest Centroid Classifier	123
3.3 Design of a Learning System	81	4.5 Locally Weighted Regression (LWR)	124
3.4 Introduction to Concept Learning	82	5. Regression Analysis	130
3.4.1 Representation of a Hypothesis	83	5.1 Introduction to Regression	130
3.4.2 Hypothesis Space	85	5.2 Introduction to Linearity, Correlation, and Causation	131
3.4.3 Heuristic Space Search	85	5.3 Introduction to Linear Regression	134
3.4.4 Generalization and Specialization	86	5.4 Validation of Regression Methods	138
3.4.5 Hypothesis Space Search by Find-S Algorithm	88	5.5 Multiple Linear Regression	141
3.4.6 Version Spaces	90	5.6 Polynomial Regression	142
3.5 Induction Biases	94	5.7 Logistic Regression	144
3.5.1 Bias and Variance	95	5.8 Ridge, Lasso, and Elastic Net Regression	147
3.5.2 Bias vs Variance Tradeoff	96	5.8.1 Ridge Regularization	148
3.5.3 Best Fit in Machine Learning	97	5.8.2 LASSO	149
3.6 Modelling in Machine Learning	97	5.8.3 Elastic Net	149
3.6.1 Model Selection and Model Evaluation	98	6. Decision Tree Learning	155
3.6.2 Re-sampling Methods	99	6.1 Introduction to Decision Tree Learning Model	155
3.7 Learning Frameworks	104	6.1.1 Structure of a Decision Tree	156
3.7.1 PAC Framework	104	6.1.2 Fundamentals of Entropy	159
3.7.2 Estimating Hypothesis Accuracy	106	6.2 Decision Tree Induction Algorithms	161
3.7.3 Hoeffding's Inequality	106	6.2.1 ID3 Tree Construction	161
3.7.4 Vapnik-Chervonenkis Dimension	107	6.2.2 C4.5 Construction	167
4. Similarity-based Learning	115	6.2.3 Classification and Regression Trees Construction	175
4.1 Introduction to Similarity or Instance-based Learning	116	6.2.4 Regression Trees	185
4.1.1 Differences Between Instance- and Model-based Learning	116	6.3 Validating and Pruning of Decision Trees	190

7. Rule-based Learning	196	9. Probabilistic Graphical Models	253
7.1 Introduction	196	9.1 Introduction	253
7.2 Sequential Covering Algorithm	198	9.2 Bayesian Belief Network	254
7.2.1 PRISM	198	9.2.1 Constructing BBN	254
7.3 First Order Rule Learning	206	9.2.2 Bayesian Inferences	256
7.3.1 FOIL (First Order Inductive Learner Algorithm)	208	9.3 Markov Chain	261
7.4 Induction as Inverted Deduction	215	9.3.1 Markov Model	261
7.5 Inverting Resolution	215	9.3.2 Hidden Markov Model	263
7.5.1 Resolution Operator (Propositional Form)	215	9.4 Problems Solved with HMM	264
7.5.2 Inverse Resolution Operator (Propositional Form)	216	9.4.1 Evaluation Problem	265
7.5.3 First Order Resolution	216	9.4.2 Computing Likelihood Probability	267
7.5.4 Inverting First Order Resolution	216	9.4.3 Decoding Problem	269
7.6 Analytical Learning or Explanation Based Learning (EBL)	217	9.4.4 Baum-Welch Algorithm	272
7.6.1 Perfect Domain Theories	218	10. Artificial Neural Networks	279
7.7 Active Learning	221	10.1 Introduction	280
7.7.1 Active Learning Mechanisms	222	10.2 Biological Neurons	280
7.7.2 Query Strategies/Selection Strategies	223	10.3 Artificial Neurons	281
7.8 Association Rule Mining	225	10.3.1 Simple Model of an Artificial Neuron	281
8. Bayesian Learning	234	10.3.2 Artificial Neural Network Structure	282
8.1 Introduction to Probability-based Learning	234	10.3.3 Activation Functions	282
8.2 Fundamentals of Bayes Theorem	235	10.4 Perceptron and Learning Theory	284
8.3 Classification Using Bayes Model	235	10.4.1 XOR Problem	287
8.3.1 Naïve Bayes Algorithm	237	10.4.2 Delta Learning Rule and Gradient Descent	288
8.3.2 Brute Force Bayes Algorithm	243	10.5 Types of Artificial Neural Networks	288
8.3.3 Bayes Optimal Classifier	243	10.5.1 Feed Forward Neural Network	289
8.3.4 Gibbs Algorithm	244	10.5.2 Fully Connected Neural Network	289
8.4 Naïve Bayes Algorithm for Continuous Attributes	244	10.5.3 Multi-Layer Perceptron (MLP)	289
8.5 Other Popular Types of Naive Bayes Classifiers	247	10.5.4 Feedback Neural Network	290
		10.6 Learning in a Multi-Layer Perceptron	290

10.7 Radial Basis Function Neural Network	297	12.3.2 Cascading	347
10.8 Self-Organizing Feature Map	301	12.4 Sequential Ensemble Models	347
10.9 Popular Applications of Artificial Neural Networks	306	12.4.1 AdaBoost	347
10.10 Advantages and Disadvantages of ANN	306	13. Clustering Algorithms	361
10.11 Challenges of Artificial Neural Networks	307	13.1 Introduction to Clustering Approaches	361
11. Support Vector Machines	312	13.2 Proximity Measures	364
11.1 Introduction to Support Vector Machines	312	13.3 Hierarchical Clustering Algorithms	368
11.2 Optimal Hyperplane	314	13.3.1 Single Linkage or MIN Algorithm	368
11.3 Functional and Geometric Margin	316	13.3.2 Complete Linkage or MAX or Clique	371
11.4 Hard Margin SVM as an Optimization Problem	319	13.3.3 Average Linkage	371
11.4.1 Lagrangian Optimization Problem	320	13.3.4 Mean-Shift Clustering Algorithm	372
11.5 Soft Margin Support Vector Machines	323	13.4 Partitional Clustering Algorithm	373
11.6 Introduction to Kernels and Non-Linear SVM	326	13.5 Density-based Methods	376
11.7 Kernel-based Non-Linear Classifier	330	13.6 Grid-based Approach	377
11.8 Support Vector Regression	331	13.7 Probability Model-based Methods	379
11.8.1 Relevance Vector Machines	333	13.7.1 Fuzzy Clustering	379
12. Ensemble Learning	339	13.7.2 Expectation-Maximization (EM) Algorithm	380
12.1 Introduction	339	13.8 Cluster Evaluation Methods	382
12.1.1 Ensembling Techniques	341	14. Reinforcement Learning	389
12.2 Parallel Ensemble Models	341	14.1 Overview of Reinforcement Learning	389
12.2.1 Voting	341	14.2 Scope of Reinforcement Learning	390
12.2.2 Bootstrap Resampling	341	14.3 Reinforcement Learning As Machine Learning	392
12.2.3 Bagging	342	14.4 Components of Reinforcement Learning	393
12.2.4 Random Forest	342	14.5 Markov Decision Process	396
12.3 Incremental Ensemble Models	346	14.6 Multi-Arm Bandit Problem and Reinforcement Problem Types	398
12.3.1 Stacking	347		

14.7 Model-based Learning (Passive Learning)	402	15.6 Evolutionary Computing	433
14.8 Model Free Methods	406	15.6.1 Simulated Annealing	433
14.8.1 Monte-Carlo Methods	407	15.6.2 Genetic Programming	434
14.8.2 Temporal Difference Learning	408	16. Deep Learning	439
14.9 Q-Learning	409	16.1 Introduction to Deep Neural Networks	439
14.10 SARSA Learning	410	16.2 Introduction to Loss Functions and Optimization	440
15. Genetic Algorithms	417	16.3 Regularization Methods	442
15.1 Overview of Genetic Algorithms	417	16.4 Convolutional Neural Networks	444
15.2 Optimization Problems and Search Spaces	419	16.5 Transfer Learning	451
15.3 General Structure of a Genetic Algorithm	420	16.6 Applications of Deep Learning	451
15.4 Genetic Algorithm Components	422	16.6.1 Robotic Control	451
15.4.1 Encoding Methods	422	16.6.2 Linear Systems and Non-linear Dynamics	452
15.4.2 Population Initialization	424	16.6.3 Data Mining	452
15.4.3 Fitness Functions	424	16.6.4 Autonomous Navigation	453
15.4.4 Selection Methods	425	16.6.5 Bioinformatics	453
15.4.5 Crossover Methods	428	16.6.6 Speech Recognition	453
15.4.6 Mutation Methods	429	16.6.7 Text Analysis	454
15.5 Case Studies in Genetic Algorithms	430	16.7 Recurrent Neural Networks	454
15.5.1 Maximization of a Function	430	16.8 LSTM and GRU	457
15.5.2 Genetic Algorithm Classifier	433	<i>Bibliography</i>	463
		<i>Index</i>	472
		<i>About the Authors</i>	480
		<i>Related Titles</i>	481

Scan for 'Appendix 1 - Python Language Fundamentals'



Scan for 'Appendix 2 - Python Packages'



Scan for 'Appendix 3 - Lab Manual with 25 Exercises'



































































































































